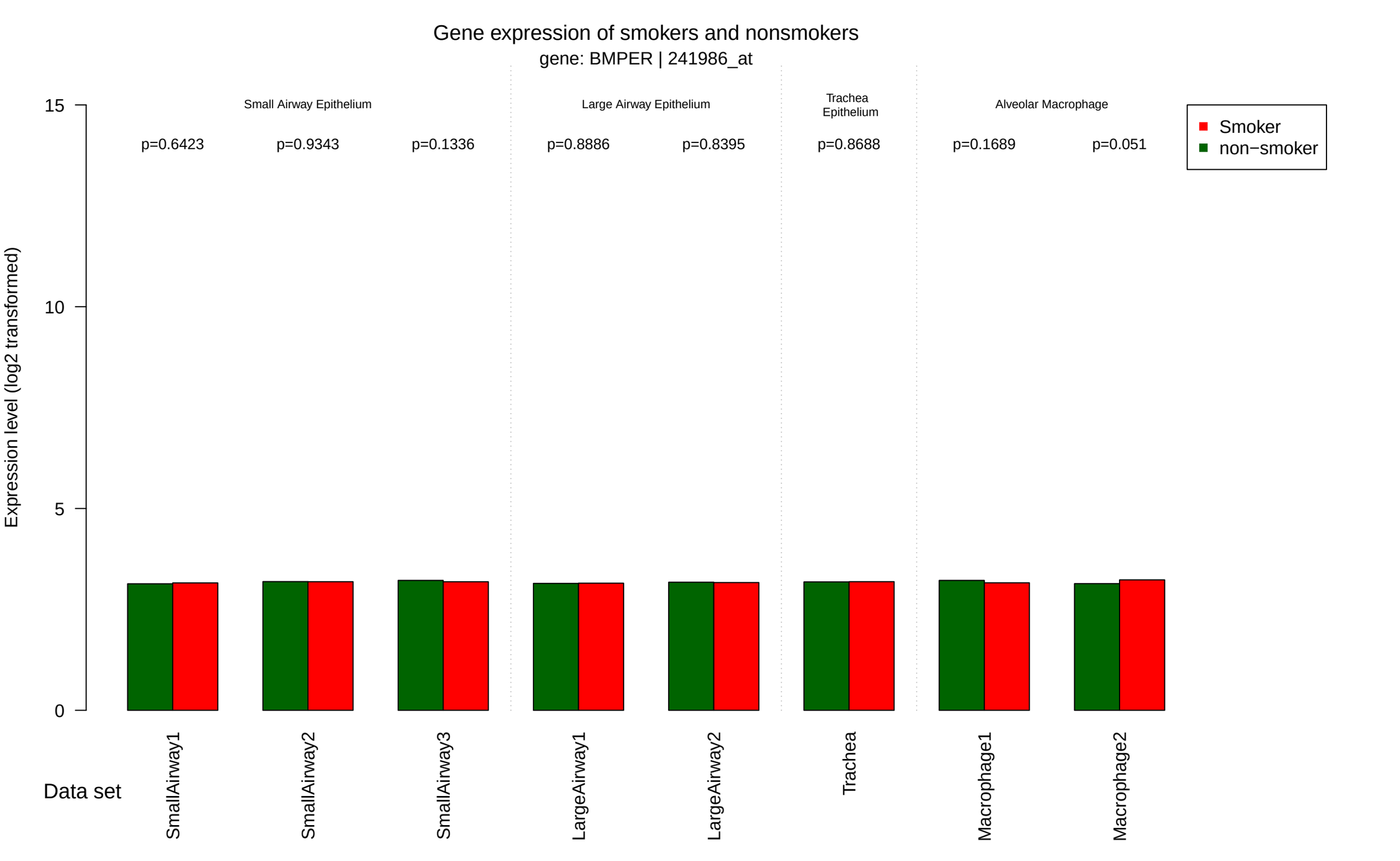
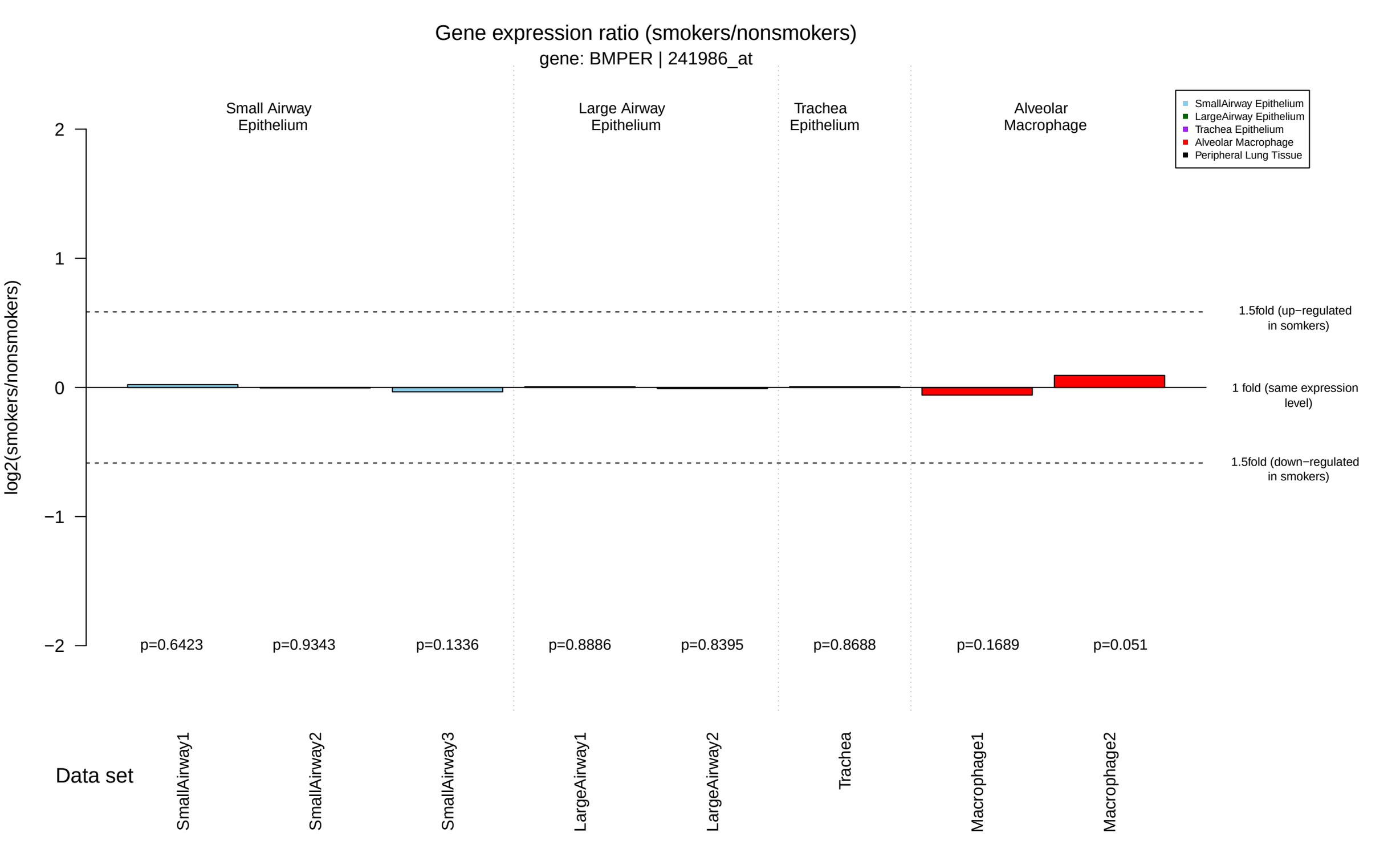
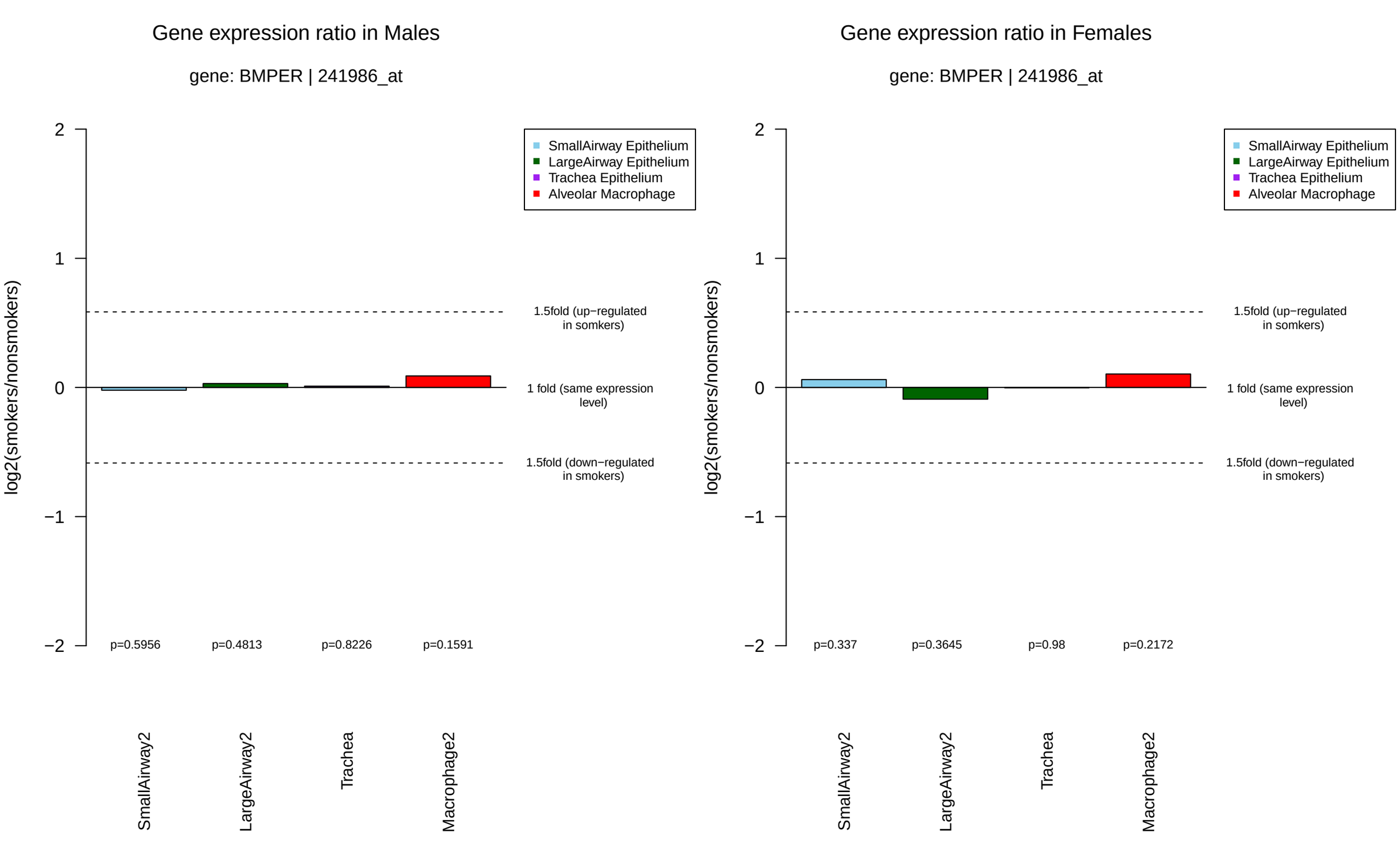
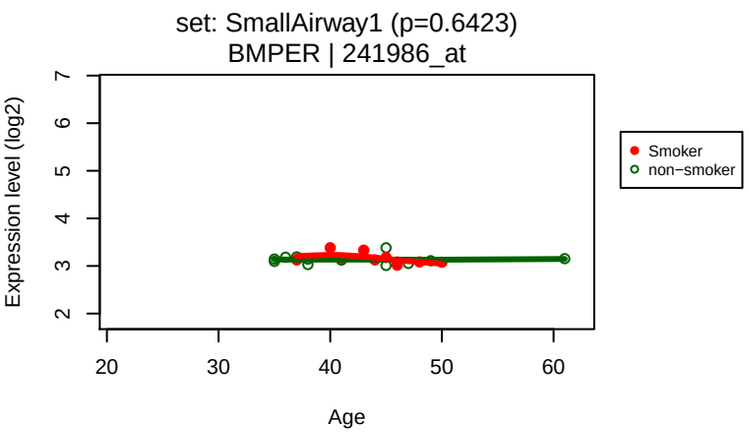
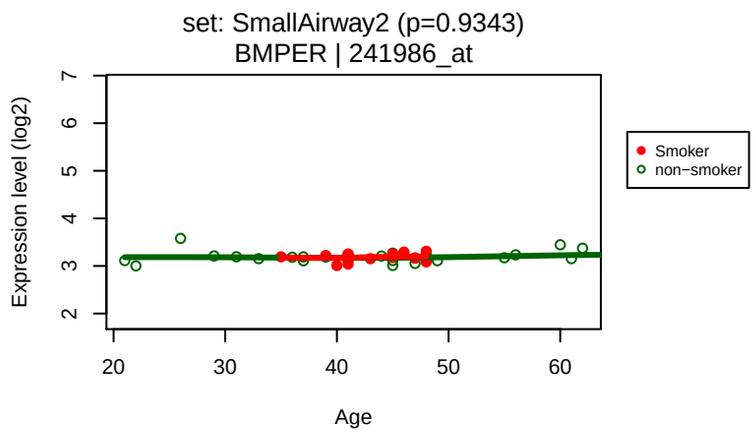
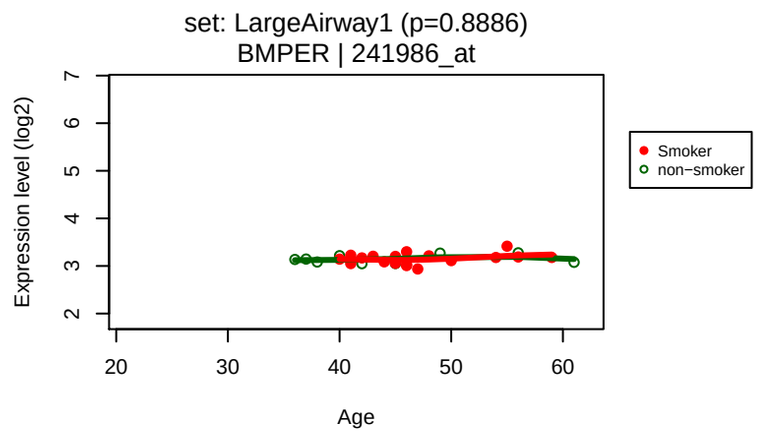

Gene expression of smokers and nonsmokers
gene: BMPER | 241986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6423
p=0.9343
p=0.1336
p=0.8886
p=0.8395
p=0.8688
p=0.1689
p=0.051
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

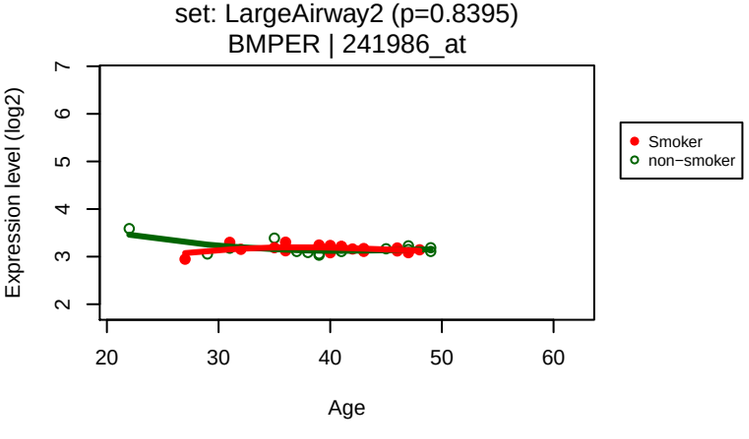

Gene expression ratio (smokers/nonsmokers)
gene: BMPER | 241986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6423
p=0.9343
p=0.1336
p=0.8886
p=0.8395
p=0.8688
p=0.1689
p=0.051
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

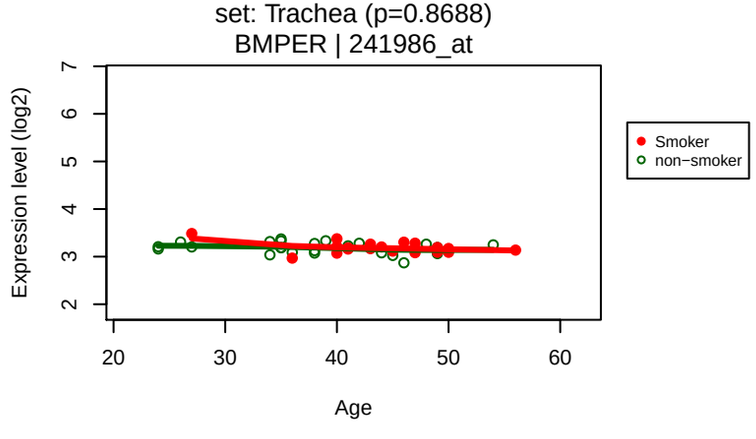

Gene expression ratio in Males
gene: BMPER | 241986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5956
p=0.4813
p=0.8226
p=0.1591
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMPER | 241986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.337
p=0.3645
p=0.98
p=0.2172
SmallAirway2
LargeAirway2
Trachea
Macrophage2

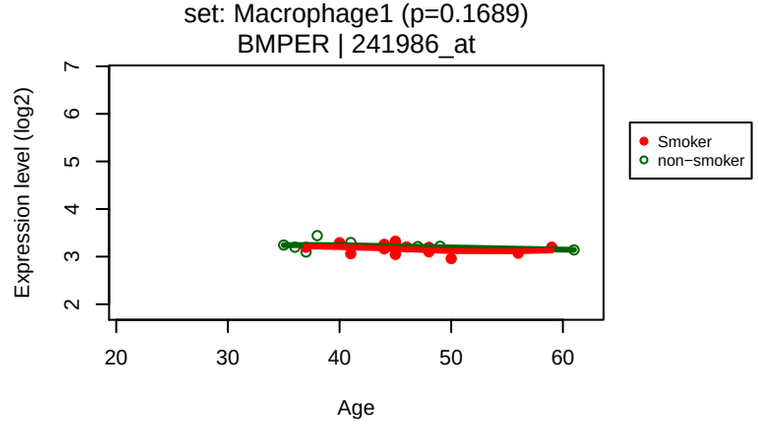

set: SmallAirway1 (p=0.6423)
BMPER | 241986_at
Expression level (log2)
Age
Smoker
non−smoker

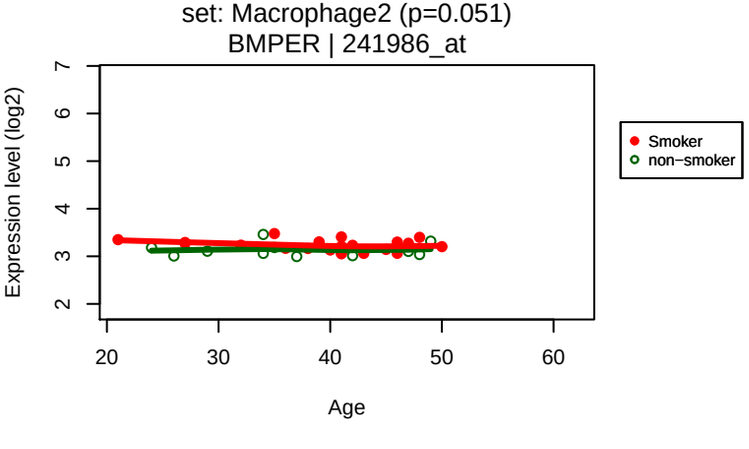

set: SmallAirway2 (p=0.9343)
BMPER | 241986_at
Expression level (log2)
Age
Smoker
non−smoker

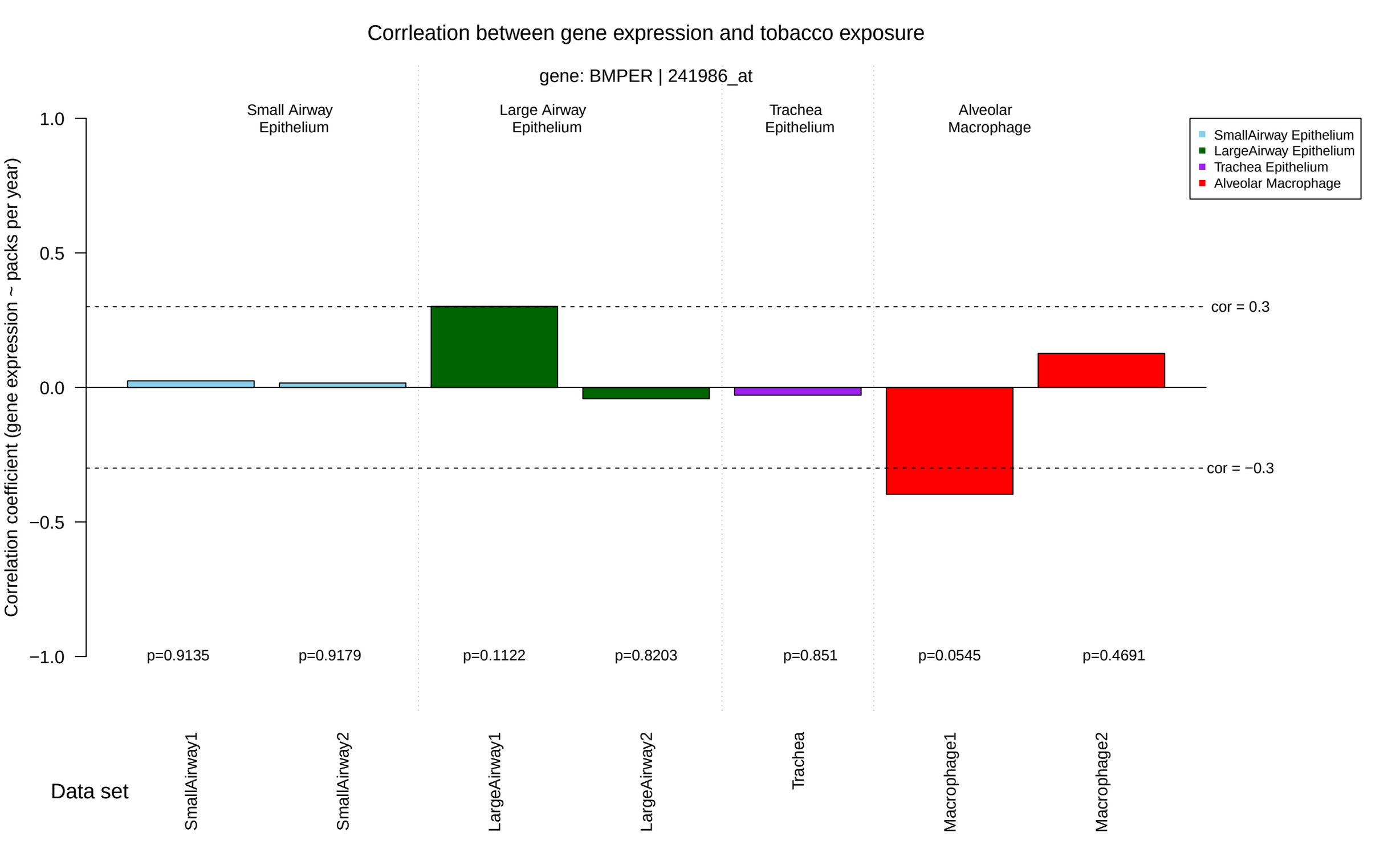

set: LargeAirway1 (p=0.8886)
BMPER | 241986_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8395)
BMPER | 241986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8688)
BMPER | 241986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1689)
BMPER | 241986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.051)
BMPER | 241986_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMPER | 241986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9135
p=0.9179
p=0.1122
p=0.8203
p=0.851
p=0.0545
p=0.4691
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2